THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **A Square and a Cube** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of perfect squares between 200 and 900 (both excluded) is: 1

(a) 14 **(b)** 17 **(c)** 15 **(d)** 16

2. The smallest number to be added to 1234 to make it a perfect square is: 1

(a) 62 **(b)** 63 **(c)** 61 **(d)** 64

3. The largest number to be subtracted from 2577 to make it a perfect square is: 1

(a) 79 **(b)** 77 **(c)** 80 **(d)** 78

4. The difference of the squares of two consecutive natural numbers is 59. The numbers are: 1

(a) 28 and 29 **(b)** 29 and 30 **(c)** 30 and 31 **(d)** 29 and 31

5. The number of digits in the square root of the 5-digit perfect square is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

6. $\sqrt[3]{-729}$ is: 1

(a) -9 **(b)** 10 **(c)** -10 **(d)** 9

7. The ones digit of 127^3 is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 3

8. If $\sqrt{x} + \sqrt{y}$ where $x = 196$ and $y = 361$, the value is: 1

(a) 34 **(b)** 266 **(c)** 5 **(d)** 33

9. The number of perfect squares between 1000 and 3000 (both excluded) is: 1

(a) 23 **(b)** 24 **(c)** 25 **(d)** 22

10. The smallest number to be added to 2345 to make it a perfect square is: 1

(a) 57 **(b)** 56 **(c)** 55 **(d)** 58

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the smallest number to be added to 3900 to make it a perfect square. What is the square root of the result? 2

12. The difference between the squares of two consecutive numbers is 75. Find the numbers. 2

13. Find the number of non-square numbers between 6^2 and 9^2. 2

14. Find the smallest number to be added to 5680 to make it a perfect square. What is the square root of the result? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A square garden has area 1225 m². Fencing costs ₹40 per metre. Find the total cost of fencing it. 3

16. A solid cube of side 6 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? 3

17. A rectangular hall 36 m by 48 m is to be paved with equal square tiles, the largest possible. Find the side of each tile and the number of tiles. 3

OR

A square garden has area 2304 m². Fencing costs ₹35 per metre. Find the total cost of fencing it. 3

18. A solid cube of side 7 cm is cut into small cubes of side 1 cm. How many small cubes are formed? How many have exactly three painted faces if the big cube was painted all over? 3

OR

A rectangular hall 42 m by 56 m is to be paved with equal square tiles, the largest possible. Find the side of each tile and the number of tiles. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the smallest square number that is divisible by each of 6, 8 and 12. 5

OR

Find the smallest cube number that is divisible by each of 4, 6 and 9. 5

20. Find the smallest square number that is divisible by each of 4, 9 and 15. 5

OR

Find the smallest cube number that is divisible by each of 6, 10 and 15. 5

END OF QUESTION PAPER Total: 40 Marks

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